

MS Module 4 Theory

Sampling

- Sampling is a process of selecting subset of observations/records from a population to make inference about different population parameters such as mean, proportion, standard deviation, etc
- It is an important step in inferential statistics since an incorrect sample may lead to wrong inference about the population

Population: set of all elements about which we wish to draw conclusions.

- Complete set of numerical information on a particular quantity in which an analyst are interested.
- Eg. impact of sales in a mart

Why do we take samples?

- Practicality
- Cost concerns
- Resources
- Time

Sampling

- Sampling is a process of selecting subset of observations/records from a population to make inference about different population parameters such as mean, proportion, standard deviation, etc
- It is an important step in inferential statistics since an incorrect sample may lead to wrong inference about the population
- Sampling is that part of statistical practice concerned with the selection of an unbiased or random subset of individual observations within a population of individuals intended to yield some knowledge about the population of concern, especially for the purposes of making predictions based on statistical inference.
- Sampling is an important aspect of data collection.

When is sampling unnecessary?

- The number of things we want to sample is small
- Data is easily accessible
- Data quality is unaffected by the number of things we look at

Example: You are interested in the relationship between team batting average and winning percentage of major league baseball teams

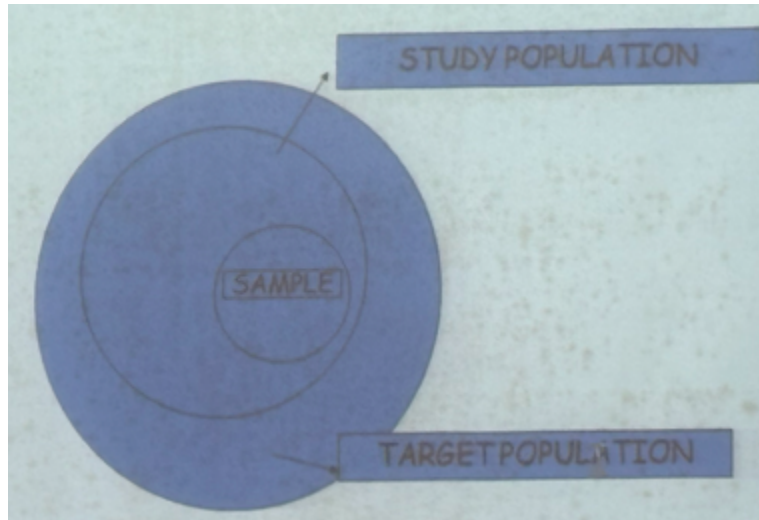
- There are only 30 major league teams
- Data on team batting averages and winning percentages are readily available

Population

- The group of elements from which a researcher samples and to which she or he might like to generalize

Sample

- A number of individual cases drawn from a larger population

**Probability versus Non Probability Sampling****Probability Sampling**

- A sample drawn in a way to give every member of the population a known (nonzero) chance of inclusion
- Probability samples are usually more representative than nonprobability samples of the populations from which they are drawn

Nonprobability Sampling

- A sample that has been drawn in a way that doesn't give every member of the population a known chance of being selected

Types of Probability Sampling

- Simple Random Sampling
- Systematic Sampling
- Stratified Sampling
- Cluster Sampling
- Multistage Sampling

Simple Random Sampling

- A probability sample in which every member of a study population has been given an equal chance of selection
 - One way to draw a simple random sample, is to put all possibilities on paper, cut them up, and then draw a sample from a hat

Systematic Sampling

- A probability sampling procedure that involves selecting every kth element from a list of population elements, after the first element has been randomly selected

Example:

- Divide the total number of elements by the number you want in your sample $24/6 = 4$
- Randomly select a number between 1 and 4 and then select every 4th element from that number

Cluster Sampling

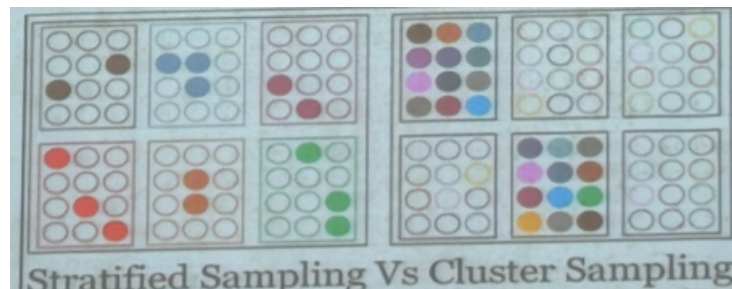
- Population is naturally divided into groups (which we call clusters).
- A random number of clusters are selected.
- Members of the selected clusters are then included in the sample

Multistage Sampling

- A probability sampling procedure that involves several stages, such as randomly selecting clusters from a population, then randomly selecting elements from each of the clusters

Stratified vs Cluster Sampling

- Stratified sampling and cluster sampling are similar; the major difference is that in a stratified sample, all strata will be represented in the sample, whereas in a cluster sampling, not all clusters will be represented



Types of Non Probability Sampling:

- Purposive sampling
- Quota sampling
- Snowball sampling
- Convenience Sampling

Purposive Sampling or Judgement sampling:

- A nonprobability sampling procedure that involves selecting elements based on a researcher's judgement about which elements will facilitate his/her investigation
- This is used primarily when there is a limited number of people that have expertise in the area being researched.

Quota Sampling

- A nonprobability sampling procedure that involves describing the target population in terms of what are thought to be relevant criteria and then selecting sample elements to represent the “relevant” subgroups in proportion to their presence in the target population
- For example, an interviewer may be told to sample 200 females and 300 males between the age of 45 and 60.
- It is this second step which makes the technique one of non-probability sampling

Snowball Sampling:

- A nonprobability sampling technique that involves using members of the group of interest to identify other members of the population.

Convenience Sampling

- A nonprobability sampling procedure that involves selecting elements that are readily accessible to the researcher
- For example, if the interviewer was to conduct a survey at a shopping center early in the morning on a given day, the people that he/she could interview would be limited to those given there at that given time, which would not represent the views of other members of society in such an area, if the survey was to be conducted at different times of day and several times per week.

Choosing a Sampling Technique

- Is it desirable to sample at all or can the whole population be used?
- Is it important to generalize to a larger population?
 - Political preference polls
- Do you have the access and ability to perform probability sampling?
- Major considerations
 - Methods
 - Theory
 - Practicality
 - Ethics

Shuffling is the process of randomly rearranging the order of items in a list or dataset to ensure that their sequence is unpredictable and unbiased.

Birth-Death Processes: Overview

A **birth-death process** is a specific type of continuous-time Markov chain where transitions can only occur between neighboring states, representing "births" (increasing the state by 1) or "deaths" (decreasing the state by 1). The state often models the size of a population or queue, where each state corresponds to a non-negative integer value [1](#) [2](#) [6](#) [9](#).

Key Features:

- **State Space:** Typically $\{0, 1, 2, \dots, N\}$ for a finite process, or $\{0, 1, 2, \dots\}$ for an infinite process [1](#).
- **Transitions:** Only possible to adjacent states: from i to $i + 1$ (birth) or $i - 1$ (death) [1](#) [2](#).
- **Transition Rates:** Each state i has a birth rate λ_i and a death rate μ_i , which may be state-dependent. If these rates are constant across states, the process is called homogeneous [1](#).
- **Markov Property:** The future evolution depends only on the current state, not the past history [1](#) [2](#).
- **Boundary Conditions:** At the minimum (e.g., state 0), deaths may be impossible ($\mu_0 = 0$); at the maximum (for finite chains), births may be impossible ($\lambda_N = 0$) [1](#).

Special Cases:

- **Pure Birth Process:** $\mu_i = 0$ for all i .
- **Pure Death Process:** $\lambda_i = 0$ for all i .
- **Poisson Process:** A pure birth process with constant birth rate [1](#).

Finite-State Markov Chains

A **finite-state Markov chain** is a Markov process with a finite set of possible states. In the context of birth-death processes:

- The chain is restricted to a finite set $\{0, 1, \dots, N\}$.
- The transition matrix (for discrete time) or infinitesimal generator (for continuous time) describes the rates or probabilities of moving between states [4](#).
- The **Chapman-Kolmogorov equations** and **Kolmogorov forward/backward equations** govern the evolution of state probabilities over time [4](#).

Long-Term Behavior:

- The existence of a limiting (stationary) distribution depends on the balance between birth and death rates [4](#).
- For example, if $\lambda < \mu$ (birth rate less than death rate), the process tends to return to lower states, possibly stabilizing at 0 [4](#).